Executive Summary

Millions of Americans suffer from diabetes, a chronic condition in which the body cannot properly produce or use insulin, often leading to high blood sugar levels. Yet, diabetes can be managed through medication, exercise, and a balanced diet. While diabetes can be managed through medication, physical activity, and diet, poor management can lead to frequent hospitalizations. These repeated readmissions not only indicate gaps in patient support but also contribute to substantial healthcare costs. This study focuses on predicting 30-day readmissions among diabetic patients and identifying which factors best signal a higher risk of returning to the hospital shortly after discharge. Because failing to detect a true readmission is more costly to the hospital than issuing a false alarm, another goal of this analysis is to develop a prediction rule that incorporates this imbalance by using a cost-adjusted decision threshold. Together, these insights can help hospitals take earlier action, improve patient outcomes, and reduce the financial burden associated with preventable readmissions.

The data used in this study comes from a large multi-hospital dataset on diabetic patients in the United States, originally compiled by researchers at Virginia Commonwealth University. Each entry represents a hospital admission and includes demographic, clinical, and diagnostic information used to explore readmission patterns.

To address our goals, we built three types of predictive models: logistic regression, LASSO, and a classification tree. Logistic regression estimates the chance that a patient will be readmitted. LASSO performs the same task while also identifying which predictors are most influential. A classification tree resembles a flowchart and makes a series of yes/no decisions to reach a prediction. We evaluated all models using AUC, a measure of how well a model separates patients who will be readmitted from those who will not; higher values indicate better separation. We trained the models on a portion of the data, tested them on unseen data, and used a final validation set to obtain an honest performance assessment. We also computed the optimal decision threshold that minimizes costs given the assumption that missing a readmission is twice as costly as predicting one that does not occur. Our results showed that LASSO logistic regression performed best overall and identified the most influential predictors, including the number of inpatient visits in the prior year and the patient's age group. These factors were among the strongest indicators of 30-day readmission risk.

While these findings provide helpful guidance for hospitals seeking to reduce preventable readmissions, several limitations should be noted. The dataset is highly imbalanced, with far fewer readmitted patients than non-readmitted ones, which makes prediction more challenging. Model performance was moderate, with AUC values around 0.65, indicating that predictions are only mostly accurate. Additionally, these models capture correlations rather than causal relationships. Finally, the data covers the years 1999–2008 and may not fully reflect current medical practices or policy environments.

Data Summary/EDA

The data used in this study comes from the Center for Clinical and Translational Research at Virginia Commonwealth University and includes information on diabetic patients across 130 U.S. hospitals between 1999 and 2008. Each record represents a unique hospital admission, totaling roughly 100,000 admissions from about 70,000 patients. Approximately 11% of patients were readmitted within 30 days.

The dataset contains demographic information, clinical characteristics, past hospital usage, diagnostic codes, and medication details, and has been pre-cleaned for analysis.

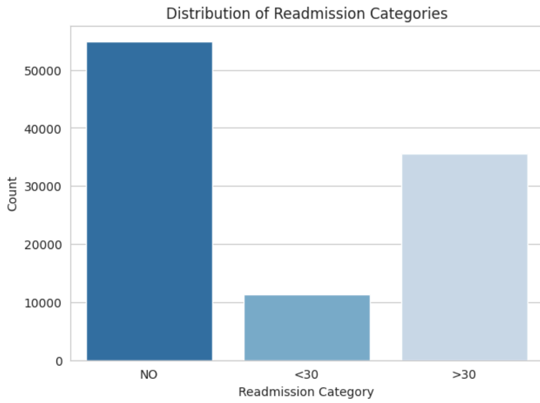

Figure 1: Distribution of Readmission

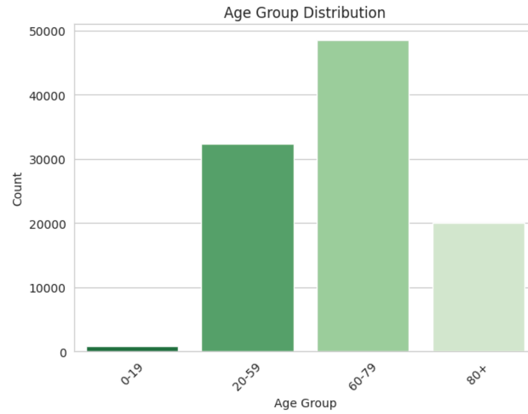

Figure 2: Distribution of Patient Age

The cleaned dataset contains 101,766 observations and 31 variables prior to dummy-encoding. After converting categorical variables into indicator (“dummy”) variables, the dataframe expands to 123 predictors, reflecting the many categorical variables in the dataset. Several continuous predictors provide information about patients’ prior and current hospital use. For example: *time_in_hospital* ranges from 1 to 14 days (median 4 days), *num_lab_procedures* has a median of 44 procedures per encounter, *number_inpatient*, *number_emergency*, and *number_outpatient* are all highly right-skewed, with most patients having zero visits in the prior year and a small number having many, and *number_diagnoses* ranges from 1 to 16 diagnoses per encounter (median 8). The outcome of interest, being whether a patient was readmitted within 30 days, is highly imbalanced: approximately 11% of encounters resulted in a <30-day readmission, while 89% did not. This imbalance motivated the use of and cost-sensitive classification later in the analysis (Appendix B).

Some issues identified during EDA were missing values, little variability in some features, and the outcome variable having three categories and needing to be shifted to a binary variable. There were numerous missing values in the *A1Credult* and *max_glu_serum*, which were filled with “None.” Some of the features that had significantly little variability were removed during preprocessing and cleaning. After dropping irrelevant identifiers (*encounter_id*, *patient_nbr*), all remaining variables were used as predictors (Appendix A). These included: Demographics: race, gender, age group; Clinical history: number_inpatient, number_emergency, number_outpatient, number_diagnoses; Lab values: max_glu_serum, A1Cresult; Diagnosis codes: diag1_mod, diag2_mod, diag3_mod; Medications: metformin, glipizide, insulin, rosiglitazone, and others; Hospital visit details: admission source, admission type, discharge disposition. Categorical variables were one-hot encoded before modeling, and continuous variables were standardized for models requiring scaling (logistic regression and LASSO).

Analyses

To build and evaluate our predictive models, we split the data into training (60%), testing (20%), and validation (20%) sets. The training set was used to fit the models, the testing set was used to compare models and select a final one based on AUC performance, and the validation set was reserved for a single, honest final evaluation. We trained three models: a standard logistic regression, a LASSO logistic regression, and a decision tree classifier. Logistic regression predicts the chance of readmission using all available features; LASSO adds an L1 penalty that shrinks less important coefficients to zero, making it especially effective in high-dimensional settings with many categorical predictors. The decision tree model makes a sequence of "if-then-else" decisions to classify patients, providing interpretability even if predictive performance is sometimes modest. Because the dataset is highly imbalanced, as there are far fewer readmissions than non-readmissions, we evaluated all models using AUC, which measures how well a model separates the two classes and does not depend on any specific probability threshold (Appendix D).

Model	Test AUC
Logistic Regression	0.650596
LASSO Logistic Regression	0.650612
Decision Tree	0.638952

Table 1: Test-Set AUCs

The LASSO logistic regression model achieved the highest AUC, though logistic regression performed similarly. Because LASSO offers additional benefits, such as feature selection, improved interpretability, and better handling of high-dimensional dummy-encoded data, it was chosen as the final model. The decision tree, while interpretable, performed slightly worse and was not selected as the primary classifier. LASSO also allowed us to identify influential predictors of 30-day readmission. The strongest predictors (by absolute coefficient value) included the number of prior inpatient visits, patient age group, discharge disposition, several diagnosis categories, and medication patterns. Positive coefficients indicate higher predicted readmission risk relative to a baseline category, whereas negative coefficients indicate lower risk. These findings align with clinical interpretation as patients with more complex medical histories, older age, or more severe diagnoses tend to be more vulnerable to short-term readmission (Appendix C).

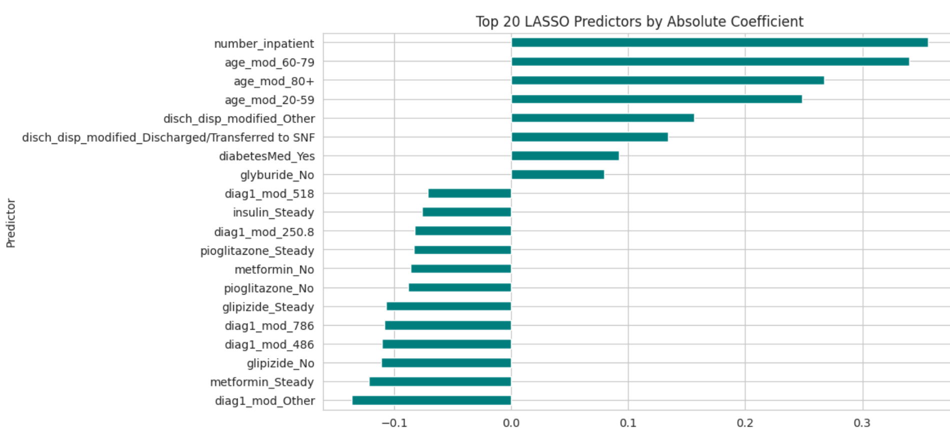

Figure 3: Top 20 LASSO Predictors by Absolute Value

Because the LASSO logistic model was the best model among the three, it was evaluated on the validation set. Initially, a standard threshold of 0.5 was used as the classification boundary for the predicted probabilities. After a model is trained, it outputs a probability score for each observation, indicating the likelihood of that observation belonging to a specific outcome (readmission or not). The threshold is then used to convert these continuous probabilities into discrete class predictions. When the threshold was set to 0.5, the validation AUC was 0.644121, the accuracy was approximately 0.65, the precision for readmission was 0.16, while the recall for readmission was 0.53. This means that the model has moderate ability to discriminate between readmission and not. At the default threshold, it misses approximately half of the true readmissions, meaning that there is a high rate of false negatives.

Knowing that a mislabeled readmission (false negative) costs double of a mislabeled non-readmission (false positive), we derive a new threshold. We calculate the optimal threshold that minimizes the cost to be 0.33 (Appendix E). Using 0.33 as the threshold, the recall for readmission rises to approximately 0.95, meaning there is a significant increase in true positives. However, the accuracy decreased to 23% due to many false positives being introduced. Yet, we know that false positives are cheaper than false negatives, so this is expected and accepted given the cost structure. Missing a true readmission may result in penalties, worse patient outcomes, and higher costs for the hospital. While extra false positives are inconvenient, they may simply just trigger additional follow-up calls or monitoring. However, when calculating the expected costs under both models, we found that at threshold 0.50, the cost would be around $8282, while the cost at threshold 0.33 would be around $15857 (Appendix F). Based on the theoretical cost ratio, the optimal threshold is 1/3. However, when evaluated on the validation set, using this threshold increased total cost because the model produced a very large number of false positives. Therefore, even though the 1/3 threshold is mathematically optimal under the assumed cost ratio, the observed data and model performance suggest that the standard 0.50 threshold actually minimizes total cost.

Conclusion

In this diabetes case study, we were able to see that LASSO logistic regression was the best model to use when predicting 30-day readmissions of patients. With an AUC around 0.65, we can say that the discrimination between readmission or not was modest but meaningful. The key predictors included prior inpatient visits, age category, discharge disposition, and diagnosis and medication patterns. Considering the 2:1 cost ratio for false negatives versus false positives, we examined both the standard 0.50 threshold and a cost-sensitive threshold of 0.33. While the lower threshold substantially improved detection of true readmissions, it resulted in higher overall cost due to an increase in false positives. The standard threshold proved more cost-effective for this dataset.

Moving forward, the LASSO model can be used to produce a risk score for each diabetes patient at discharge. Patients can be flagged with predicted probability ≥ 0.33 for enhanced discharge planning, follow-up within 7-14 days, and possible care management interventions.

When looking at our study, it is important to note that we only used historical data, and did not have any information on social determinants or physician notes to get a deeper understanding of the situations for these patients. With the model being trained on data from 1999-2008, some practice patterns, treatments, and policies may be outdated as this data is from over 10 years ago. Finally, it is necessary to mention that the model is solely predictive, not causal, meaning that the relationships found between these features and readmissions do not mean that these features cause readmittance, rather that they both happen simultaneously. Going forward, more nuanced and detailed cost estimates could be used to ensure the threshold is more precise. Additionally, testing these models with more recent data would be paramount in seeing how this model would truly benefit your hospital.

Appendix A - Data Cleaning & Preprocessing Details

```
readmission["max_glu_serum"] = readmission["max_glu_serum"].replace("None",
"None").fillna("None")
readmission["A1Cresult"] = readmission["A1Cresult"].replace("None",
"None").fillna("None")


# Binary target: 1 = readmitted within 30 days, 0 = otherwise
y = (df["readmitted"] == "<30").astype(int)

# Drop ID and original readmission column from predictors
df = df.drop(columns=["encounter_id", "patient_nbr", "readmitted"])

#Split categorical features
cat_cols = [
   'race', 'gender', 'max_glu_serum', 'A1Cresult', 'metformin',
   'glimepiride', 'glipizide', 'glyburide', 'pioglitazone',
   'rosiglitazone', 'insulin', 'change', 'diabetesMed',
   'disch_disp_modified', 'adm_src_mod', 'adm_typ_mod',
   'age_mod', 'diag1_mod', 'diag2_mod', 'diag3_mod'
]

for col in cat_cols:
   readmission[col] = readmission[col].astype('category')

# One-hot encode categorical variables
X = pd.get_dummies(df, columns=cat_cols, drop_first=True)

print("X shape after dummies:", X.shape)


# Split Data-- Train: 60%, Test: 20%, Validation: 20%
X_train, X_temp, y_train, y_temp = train_test_split(
   X, y,
   test_size=0.40,
   random_state=0,
   stratify=y
)

# Second: temp -> test (20%) and validation (20%)
X_test, X_val, y_test, y_val = train_test_split(
   X_temp, y_temp,
   test_size=0.50,     # 50% of 40% = 20% of total
   random_state=0,
```

```
   stratify=y_temp
)


scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
X_val_scaled = scaler.transform(X_val)
```

Appendix B - Summary Tables

	time_in_hospital	num_lab_procedures	num_procedures	num_medications	number_outpatient	number_emergency	number_inpatient	number_diagnoses
count	101766	101766	101766	101766	101766	101766	101766	101766
mean	4.39599	43.0956	1.33973	16.0218	0.369357	0.197836	0.635566	7.42261
std	2.98511	19.6744	1.70581	8.12757	1.26727	0.930472	1.26286	1.9336
min	1	1	0	1	0	0	0	1
25%	2	31	0	10	0	0	0	6
50%	4	44	1	15	0	0	0	8
75%	6	57	2	20	0	0	1	9
max	14	132	6	81	42	76	21	16

Table 1: Summary Statistics for Numerical Variables

race	count
Caucasian	76099
AfricanAmerican	19210
?	2273
Hispanic	2037
Other	1506
Asian	641

Table 2: Race Frequency Table

gender	count
Female	54708
Male	47055
Unknown/Invalid	3

Table 3: Gender Frequency Table

age_mod	count
60-79	48551
20-59	32373
80+	19990
0-19	852

Table 4: Age Frequency Table

diabetesMed	count
Yes	78363
No	23403

Table 5: Frequency Table of if Diabetes Meds Were Prescribed

insulin	count
No	47383
Steady	30849
Down	12218
Up	11316

Table 6: Frequency Table of Their Insulin Levels

disch_disp_modified	count
Discharged to home	60234
Other	14676
Discharged/Transferred to SNF	13954
Discharged to home with Home Health Service	12902

Table 7: Discharge Frequency Table

Appendix C - Full LASSO Coefficient Table

variable	coefficient	abs_coef	nonzero
number_inpatient	0.355996	0.355996	True
age_mod_60-79	0.340266	0.340266	True
age_mod_80+	0.267449	0.267449	True
age_mod_20-59	0.24878	0.24878	True
disch_disp_modified_Other	0.156332	0.156332	True
diag1_mod_Other	-0.136428	0.136428	True
disch_disp_modified_Discharged/Transferred to SNF	0.133848	0.133848	True
metformin_Steady	-0.121765	0.121765	True
glipizide_No	-0.111156	0.111156	True
diag1_mod_486	-0.110936	0.110936	True
diag1_mod_786	-0.108581	0.108581	True
glipizide_Steady	-0.106857	0.106857	True
diabetesMed_Yes	0.0919959	0.0919959	True
pioglitazone_No	-0.0885692	0.0885692	True
metformin_No	-0.0858511	0.0858511	True
pioglitazone_Steady	-0.0831898	0.0831898	True
diag1_mod_250.8	-0.0825353	0.0825353	True
glyburide_No	0.0794732	0.0794732	True
insulin_Steady	-0.0764982	0.0764982	True
diag1_mod_518	-0.0718533	0.0718533	True
number_emergency	0.0698949	0.0698949	True
insulin_No	-0.0688632	0.0688632	True
glimepiride_Steady	-0.0670265	0.0670265	True
diag1_mod_682	-0.0668927	0.0668927	True
disch_disp_modified_Discharged to home with Home Health Service	0.0634194	0.0634194	True
diag1_mod_435	-0.0611719	0.0611719	True
diag1_mod_599	-0.0611281	0.0611281	True
diag1_mod_427	-0.0591339	0.0591339	True
diag3_mod_250.6	0.0588402	0.0588402	True
diag3_mod_Other	0.0563892	0.0563892	True
diag2_mod_Other	0.0561758	0.0561758	True
diag2_mod_276	0.0561276	0.0561276	True
diag1_mod_414	-0.05451	0.05451	True
glyburide_Steady	0.0542942	0.0542942	True
diag1_mod_38	-0.0542792	0.0542792	True
diag1_mod_780	-0.0536084	0.0536084	True
diag3_mod_403	0.0527843	0.0527843	True
race_Caucasian	0.0494913	0.0494913	True
diag1_mod_491	-0.0475464	0.0475464	True
adm_src_mod_Transfer from Home Health	-0.0463569	0.0463569	True
adm_typ_mod_Emergency	0.0450933	0.0450933	True
diag3_mod_585	0.0441056	0.0441056	True
race_AfricanAmerican	0.0440062	0.0440062	True
diag1_mod_493	-0.0433414	0.0433414	True
diag1_mod_996	-0.0420326	0.0420326	True
A1Cresult_None	0.0420185	0.0420185	True
diag1_mod_715	-0.0419964	0.0419964	True
metformin_Up	-0.0411905	0.0411905	True
diag2_mod_491	0.0379888	0.0379888	True
diag2_mod_250.01	0.0372181	0.0372181	True
diag2_mod_403	0.0371417	0.0371417	True
diag2_mod_707	0.0366769	0.0366769	True
diag3_mod_496	0.0362879	0.0362879	True
adm_typ_mod_Urgent	0.0345133	0.0345133	True
diag1_mod_410	-0.0344354	0.0344354	True
diag2_mod_428	0.0340801	0.0340801	True
diag2_mod_411	0.033528	0.033528	True
time_in_hospital	0.0327967	0.0327967	True
rosiglitazone_No	0.0325373	0.0325373	True
diag1_mod_428	-0.0322542	0.0322542	True
gender_Unknown/Invalid	-0.0316813	0.0316813	True
glimepiride_No	-0.0313936	0.0313936	True
num_medications	0.0312847	0.0312847	True
insulin_Up	-0.0305746	0.0305746	True
rosiglitazone_Steady	0.030451	0.030451	True
diag2_mod_585	0.0288405	0.0288405	True
num_procedures	-0.0283443	0.0283443	True
diag3_mod_?	-0.0276839	0.0276839	True
diag2_mod_401	-0.0250565	0.0250565	True
diag2_mod_682	0.0249784	0.0249784	True
diag2_mod_250.02	0.0242919	0.0242919	True

pioglitazone_Up	-0.0209583	0.0209583	True
adm_src_mod_Other	-0.0204993	0.0204993	True
adm_src_mod_Physician Referral	0.0200753	0.0200753	True
A1Cresult_Norm	-0.0192327	0.0192327	True
diag1_mod_434	0.0190598	0.0190598	True
max_glu_serum_None	-0.0186794	0.0186794	True
diag3_mod_414	-0.0180647	0.0180647	True
diag3_mod_V45	-0.0180158	0.0180158	True
diag2_mod_425	0.017088	0.017088	True
diag3_mod_428	0.0170172	0.0170172	True
race_Hispanic	0.0165847	0.0165847	True
number_diagnoses	0.0156953	0.0156953	True
diag1_mod_584	-0.0156115	0.0156115	True
diag2_mod_518	-0.0151239	0.0151239	True
diag3_mod_401	-0.0149573	0.0149573	True
diag2_mod_427	0.0147624	0.0147624	True
glyburide_Up	0.0136712	0.0136712	True
diag2_mod_285	-0.0132743	0.0132743	True
race_Asian	0.0126149	0.0126149	True
diag3_mod_250.02	0.012086	0.012086	True
diag3_mod_707	0.0118299	0.0118299	True
diag3_mod_285	-0.0117025	0.0117025	True
diag2_mod_414	0.0109918	0.0109918	True
glipizide_Up	-0.0108323	0.0108323	True
diag1_mod_577	-0.0106101	0.0106101	True
diag2_mod_413	-0.0101259	0.0101259	True
diag3_mod_272	-0.0100032	0.0100032	True
diag3_mod_780	0.00943976	0.00943976	True
diag3_mod_425	0.00913417	0.00913417	True
race_Other	0.00888558	0.00888558	True
diag2_mod_780	-0.00872863	0.00872863	True
diag1_mod_276	-0.00845143	0.00845143	True
num_lab_procedures	0.00743714	0.00743714	True
change_No	0.00669965	0.00669965	True
gender_Male	-0.00635301	0.00635301	True
glimepiride_Up	-0.00583639	0.00583639	True
adm_typ_mod_Other	0.00497433	0.00497433	True
diag2_mod_584	0.00470852	0.00470852	True
max_glu_serum_>300	-0.00447004	0.00447004	True
diag3_mod_427	0.00434411	0.00434411	True
diag2_mod_599	0.00426323	0.00426323	True
diag3_mod_599	-0.00416773	0.00416773	True
diag3_mod_276	0.00372703	0.00372703	True
diag2_mod_486	-0.00364937	0.00364937	True
diag2_mod_424	-0.00351586	0.00351586	True
A1Cresult_>8	-0.00341416	0.00341416	True
number_outpatient	-0.00333336	0.00333336	True
max_glu_serum_Norm	0.00275092	0.00275092	True
rosiglitazone_Up	0.00236399	0.00236399	True
diag1_mod_820	0.00180904	0.00180904	True
diag3_mod_424	0.000295572	0.000295572	True
diag2_mod_496	0	0	False

Table 1: LASSO Features and Corresponding Coefficients

Appendix D - Model Evaluation Metrics

Logistic Regression

	precision	recall	f1-score	support
0	0.92	0.66	0.77	18082
1	0.17	0.56	0.26	2271
accuracy			0.65	20353
macro avg	0.55	0.61	0.52	20353
weighted avg	0.84	0.65	0.72	20353

ROC-AUC (Logistic): 0.6505960346697911

LASSO Regression

	precision	recall	f1-score	support
0	0.92	0.67	0.77	18082
1	0.17	0.56	0.26	2271
accuracy			0.65	20353
macro avg	0.55	0.61	0.52	20353
weighted avg	0.84	0.65	0.72	20353

ROC-AUC (LASSO): 0.6506122775198322

Decision Tree

	precision	recall	f1-score	support
0	0.91	0.84	0.87	18082
1	0.20	0.31	0.24	2271
accuracy			0.78	20353
macro avg	0.55	0.58	0.56	20353
weighted avg	0.83	0.78	0.80	20353

ROC-AUC (Logistic): 0.610617339834175

Model Comparison on TEST set (by AUC)

	Model	Test AUC
0	Logistic	0.650596
1	LASSO	0.650612
2	Decision Tree	0.610617

Final Evaluation on Validation set with LASSO (threshold = 0.5)

	precision	recall	f1-score	support
0	0.92	0.66	0.77	18082
1	0.16	0.53	0.25	2272
accuracy			0.65	20354
macro avg	0.54	0.60	0.51	20354
weighted avg	0.83	0.65	0.71	20354

Validation AUC (LASSO): 0.6441207143591556

FINAL MODEL EVALUATION (LASSO, threshold = 1/3)

	precision	recall	f1-score	support
0	0.95	0.14	0.24	18082
1	0.12	0.95	0.21	2272
accuracy			0.23	20354
macro avg	0.54	0.54	0.23	20354
weighted avg	0.86	0.23	0.24	20354

Validation AUC (LASSO): 0.6441207143591556

Appendix E - Threshold Derivation

Derivation Copied Over from Latex Overleaf

- Let Y denote 30-day readmittance:

$$Y = 1 : \text{readmitted within 30 days}, \qquad Y = 0 : \text{no readmittance}$$

- **False Negative (FN)**: predict no readmittance but true readmittance; cost $= 2$
- **False Positive (FP)**: predict readmittance but no true readmittance; cost $= 1$

$$\text{Total Cost} = 2 \cdot FN + 1 \cdot FP$$

Let

$$\hat{p}(x) = P(Y = 1 \mid X = x).$$

We predict readmittance ($\hat{Y} = 1$) when:

$$FP \cdot P(Y = 0 \mid x) \leq FN \cdot P(Y = 1 \mid x).$$

Substitute $P(Y = 1 \mid x) = \hat{p}(x)$ and $P(Y = 0 \mid x) = 1 - \hat{p}(x)$:

$$FP(1 - \hat{p}(x)) \leq FN\,\hat{p}(x)$$

$$FP - FP\,\hat{p}(x) \leq FN\,\hat{p}(x)$$

$$FP \leq (FP + FN)\,\hat{p}(x)$$

Solve for $\hat{p}(x)$:

$$\hat{p}(x) \geq \frac{FP}{FP + FN}.$$

With $FP = 1$ and $FN = 2$:

$$\hat{p}(x) \geq \frac{1}{1+2} = \frac{1}{3} \approx 0.33.$$

Predict readmittance if $\hat{p}(x) \geq 0.33$.

Appendix F - Cost Calculations

```python
from sklearn.metrics import confusion_matrix

# cost weights
C_FN = 2  # cost of mislabeling a readmission
C_FP = 1  # cost of mislabeling a non-readmission

def evaluate_cost(y_true, y_proba, threshold, C_FN=2, C_FP=1):
    """
    Compute confusion matrix and total cost at a given threshold.
    Total cost = C_FN * FN + C_FP * FP
    """
    y_pred = (y_proba >= threshold).astype(int)
    tn, fp, fn, tp = confusion_matrix(y_true, y_pred).ravel()
    total_cost = (C_FN * fn) + (C_FP * fp)

    print(f"\n=== Threshold = {threshold:.2f} ===")
    print(f"TN: {tn}, FP: {fp}, FN: {fn}, TP: {tp}")
    print(f"Total cost (2*FN + 1*FP): {total_cost}")

    return total_cost

# Compare thresholds 0.50 vs 1/3
cost_05  = evaluate_cost(y_val, y_val_proba_lasso, threshold=0.50, C_FN=C_FN,
C_FP=C_FP)
cost_033 = evaluate_cost(y_val, y_val_proba_lasso, threshold=1/3,  C_FN=C_FN,
C_FP=C_FP)

print("\nSummary:")
print(f"Cost at threshold 0.50: {cost_05}")
print(f"Cost at threshold 0.33: {cost_033}")
```

```
=== Threshold = 0.50 ===
TN: 11930, FP: 6152, FN: 1065, TP: 1207
Total cost (2*FN + 1*FP): 8282

=== Threshold = 0.33 ===
TN: 2461, FP: 15621, FN: 118, TP: 2154
Total cost (2*FN + 1*FP): 15857

Summary:
Cost at threshold 0.50: 8282
Cost at threshold 0.33: 15857
```